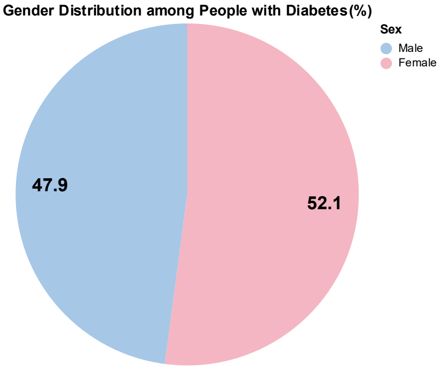
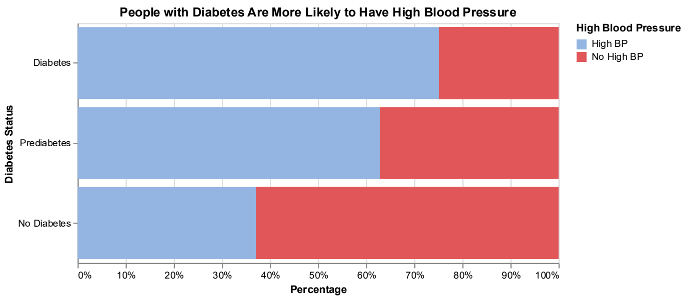
Understanding Patterns Among Individuals with Diabetes

This analysis uses data from the Diabetes Health Indicators Dataset, sourced from Kaggle, which is derived from the Centers for Disease Control and Prevention's (CDC) Behavioral Risk Factor Surveillance System (BRFSS) 2015 telephone survey of U.S. adults. The dataset includes self-reported health and lifestyle information such as age, BMI, blood pressure, and diabetes status. It is important to note that this analysis is observational and aims to identify patterns and associations among variables rather than establish any causal relationships.

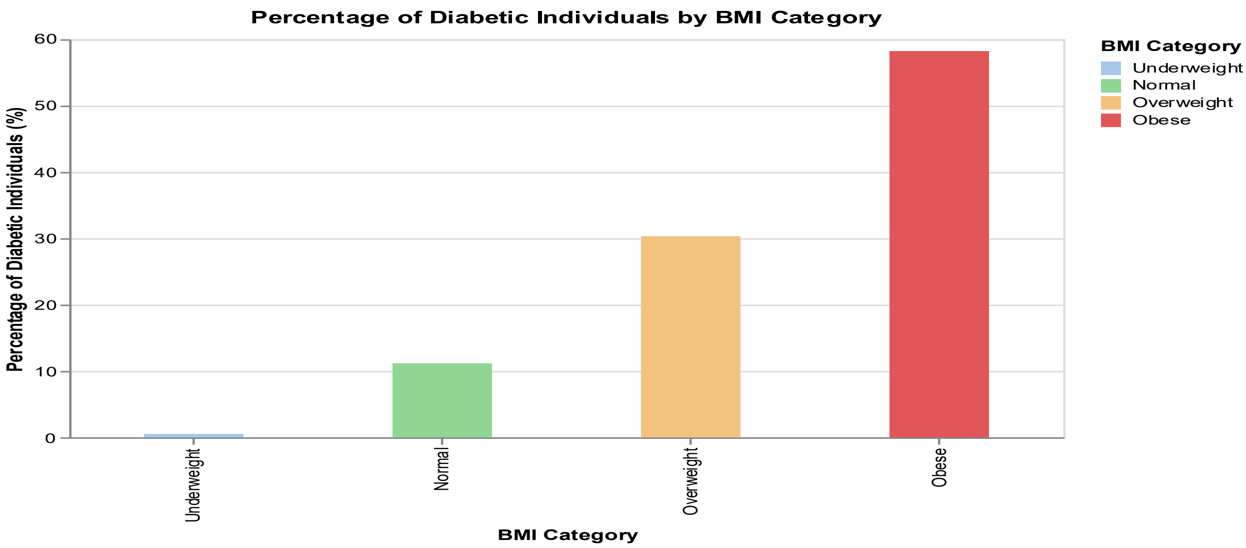


An interesting observation in our dataset is that females appear more likely to have diabetes. However, national data from the CDC (2021–2023) show higher age-adjusted prevalence among men (18.0%) than women (13.7%) (CDC, 2023). Similarly, a global review found that men are often diagnosed earlier and at lower body fat levels, with a higher prevalence of type 2 diabetes in younger and middle-aged groups (Kautzky-Willer et al., 2023).

Our dataset shows individuals with diabetes are much more likely to report high blood pressure than those without the condition. This mirrors national evidence showing that 50–80% of U.S. adults with type 2 diabetes also have hypertension (Jia & Sowers, 2021)



Our data show a clear upward trend in diabetes prevalence with increasing BMI , from almost none in underweight individuals to over half among those classified as obese. This strong association reflects the role of excess body fat in driving insulin resistance.



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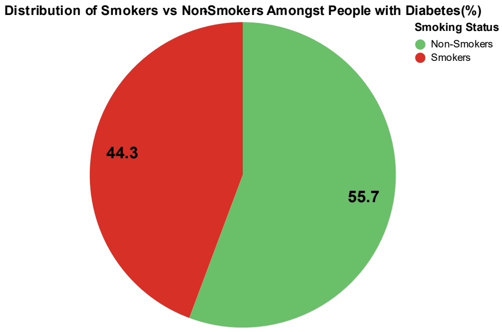
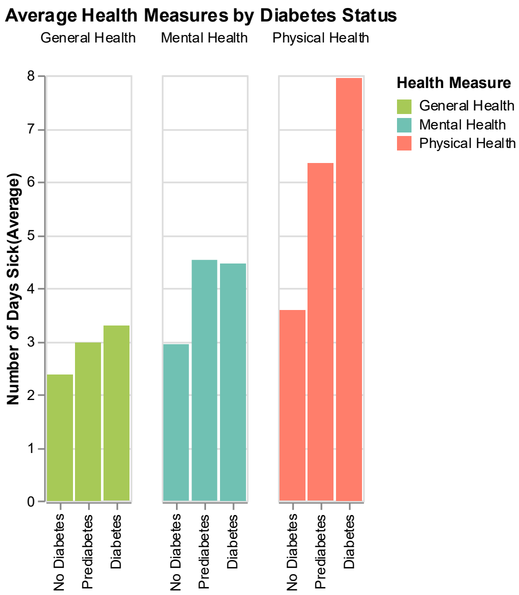
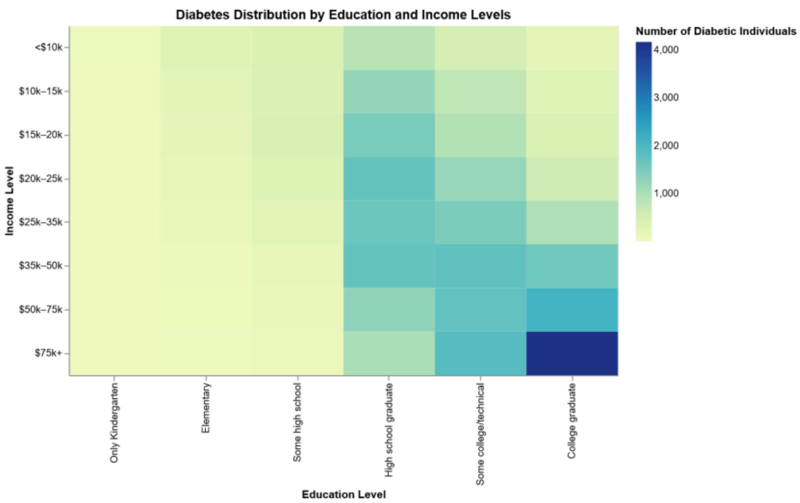
Although the percentage of smokers among individuals with diabetes (44.3%) is lower than non-smokers, it remains a substantial share that cannot be ignored. Smoking is a well-established risk factor that worsens insulin resistance and increases the likelihood of developing type 2 diabetes and its complications (CDC, 2020).





It shows that people with diabetes report more unhealthy days across all three measures, particularly in physical health, where the average nearly doubles compared to non-diabetic individuals. These findings suggest that diabetes is linked not only to physical limitations but also to broader well-being challenges.

An interesting pattern in our dataset shows more reported diabetes cases among higher-income and highly educated individuals. This contrasts with typical trends but may reflect lifestyle factors such as sedentary work, calorie-dense diets, and stress that increase risk despite better healthcare access. Supporting this, a U.S. study (2014–2018) found that adults in sedentary occupations had higher diabetes rates than those in active jobs, linking work-related inactivity to greater diabetes prevalence in higher-income groups (Aravindalochanan, Kumpatla, Rengarajan, Rajan, & Viswanathan, 2014).



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